Project Description

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Setup of the Original Model as in @Agu2023

We implement the dynamic discrete-choice demand model of @Agu2023. A consumer i each period t chooses $y_{it} \in \{0, 1, \dots, J\}$ where $y_{it} = 0$ is "no purchase" and $y_{it} = j > 0$ is buying brand j. The endogenous state $x_{it} = (\ell_{it}, d_{it})$ records the last brand purchased and the duration since the last purchase. Prices have a persistent component z_{it} and a transitory (promotion) component e_{it}. With i.i.d. Type-I Extreme Value shocks the per-period payoff absent the shock is

$$u_{\alpha_i}(y, x, p) = \begin{cases} \alpha_i(\ell) + \gamma h(\mu_i) - \beta^{dep}(\ell)\, d & y = 0, \\ \alpha_i(j) + \gamma h(\mu_i - p(j)) - \beta^{sc}(\ell, j) & y = j > 0. \end{cases}$$

The block below sets up the primitives, solves the consumer's Bellman equation by VFI (used only for simulating data), simulates a panel, and estimates $\theta = (\gamma, \beta^{sc}, \beta^{dep})$ by the sufficient-statistic Conditional Maximum Likelihood (CML) estimator described in Sections 3–4 of the paper.

Table 1: Simulation and CML estimation summary.

Statistic	Value
Consumers	30000
Periods	52
Brands	2
Observed purchases	563464 of 1560000
Purchase rate	36.1%
Sufficient-statistic cells with $e_3 = e_4$	573
Price-contrast std. dev.	10.140
Nonzero price-contrast share	55.7%
Alternating-history share	16.6%
Depreciation spell cells	8557
Depreciation duration cap d^*	2
Minimum T for depreciation CML	5
Constant-promotion spell restriction	True
Alpha FE sample std. dev.	0.249, 0.453
Optimizer converged?	Yes
Optimizer iterations	15
Monte Carlo enabled?	Yes
Monte Carlo replications	10
Monte Carlo consumers	30000
Monte Carlo periods	52

Table 2: CML parameter estimates.

Parameter	True	CML	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%	Bias
γ	1.035	1.643	0.850	1.932	0.053	-0.024	3.309	0.608
$\tilde{\beta}_{1,2}$	1.050	1.149	0.129	8.942	0.000	0.897	1.401	0.099

Table 3: CML depreciation estimates from no-purchase spell histories.

Parameter	True	CML	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%	Bias
β_1^{dep}	0.250	0.219	0.022	9.987	0.000	0.176	0.262	-0.031
β_2^{dep}	0.250	0.185	0.169	1.093	0.274	-0.147	0.516	-0.065

Table 4: General CML fit and test statistics.

Statistic	Value
Log-likelihood	-183.302
Null log-likelihood	-397.173
Pseudo R^2	0.538
AIC	370.603
BIC	379.305
LR test statistic	427.743
LR test df	2
LR test p-value	0.000

Table 5: Monte Carlo simulation and CML estimation summary across 10 replications.

Statistic	Value
Replications	10
Consumers per replication	30000
Periods per replication	52
Brands	2
Mean observed purchases	556890.3 of 1560000
Mean purchase rate	35.7%
Mean sufficient-statistic cells with $e_3 = e_4$	721.2
Mean price-contrast std. dev.	10.171
Mean nonzero price-contrast share	58.4%
Mean alternating-history share	16.2%
Mean depreciation spell cells	8379.0
Depreciation duration cap d^*	2
Minimum T for depreciation CML	5
Constant-promotion spell restriction	True
Mean alpha FE sample std. dev.	0.313, 0.315
Purchase optimizer convergence share	100.0%
Mean purchase optimizer iterations	15.6
Depreciation estimator availability share	100.0%
Depreciation optimizer convergence share	100.0%
Mean depreciation optimizer iterations	9.1

Table 6: Monte Carlo performance summary across 10 replications with $N = 30000$ and $T = 52$.

Parameter	True	CML	Std. err.	z-stat	p-value	CI 2.5%	CI 97.5%	Bias	Reps.	RMSE
γ	1.035	1.149	0.546	2.103	0.035	0.078	2.219	0.114	10	0.530
$\tilde{\beta}_{1,2}$	1.050	1.099	0.136	8.103	0.000	0.833	1.365	0.049	10	0.138
β_1^{dep}	0.250	0.238	0.015	16.100	0.000	0.209	0.267	-0.012	10	0.019
β_2^{dep}	0.250	0.269	0.127	2.120	0.034	0.020	0.517	0.019	10	0.122

Note: The first CML table uses purchase histories (k, j, k, j) and (k, k, j, j) to estimate γ and

$\tilde{\beta}_{k,j}$. The depreciation table uses the no-purchase spell histories from Section 3.3, imposing the simulated duration cap $d^* = 2$ as the point where the depreciation state is flat.